Contribution Journal

In this lab, I completed and debugged the full machine learning pipeline on the Titanic dataset using Python, pandas, and scikit-learn. I resolved code syntax and import issues, handled missing data, and encoded categorical features like gender into binary format for model training. I implemented, trained, and evaluated both a LinearRegression model to predict passenger fares (evaluating performance with Mean Squared Error and Root Mean Squared Error) and a LogisticRegression model to predict passenger survival (evaluating performance with Accuracy Score). Finally, I analyzed feature weights using the models' .coef_ attributes and documented key conceptual differences between continuous regression metrics and categorical classification metrics.